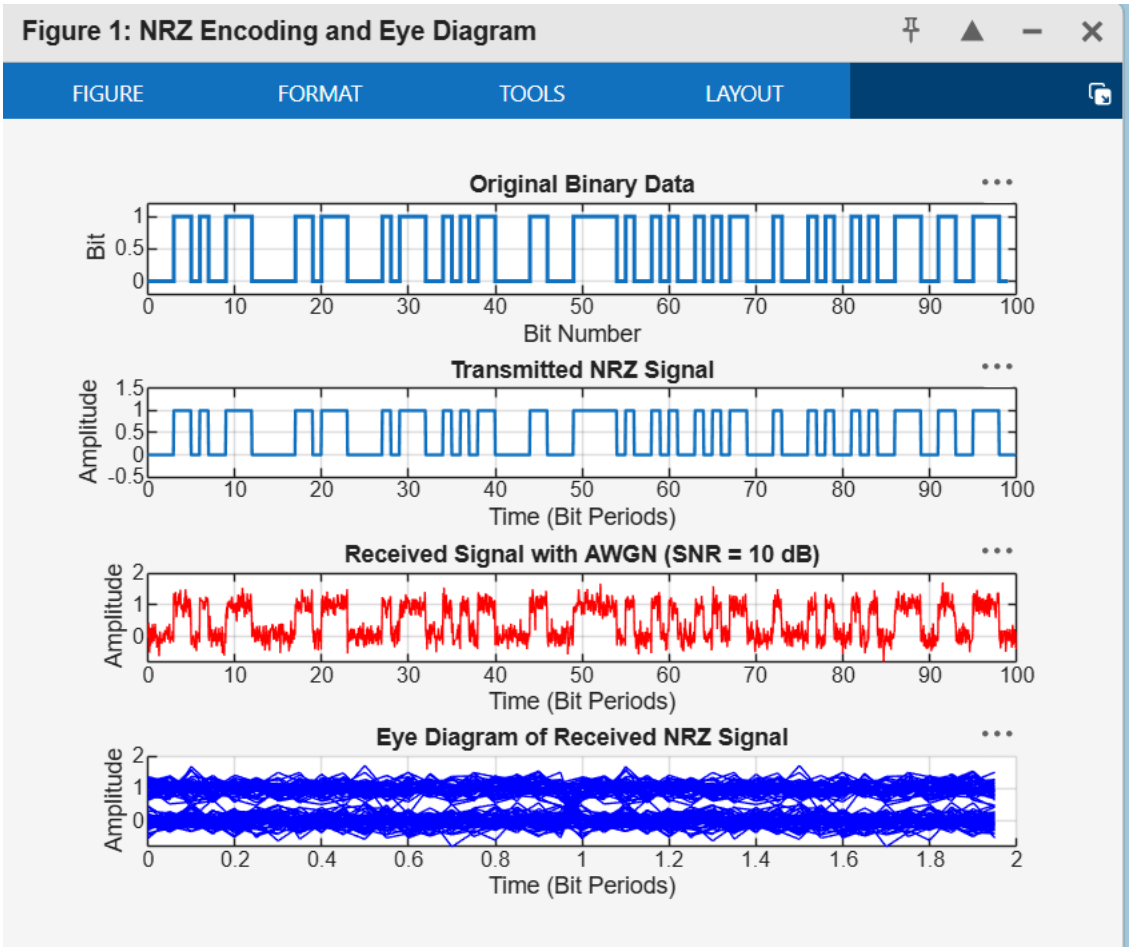
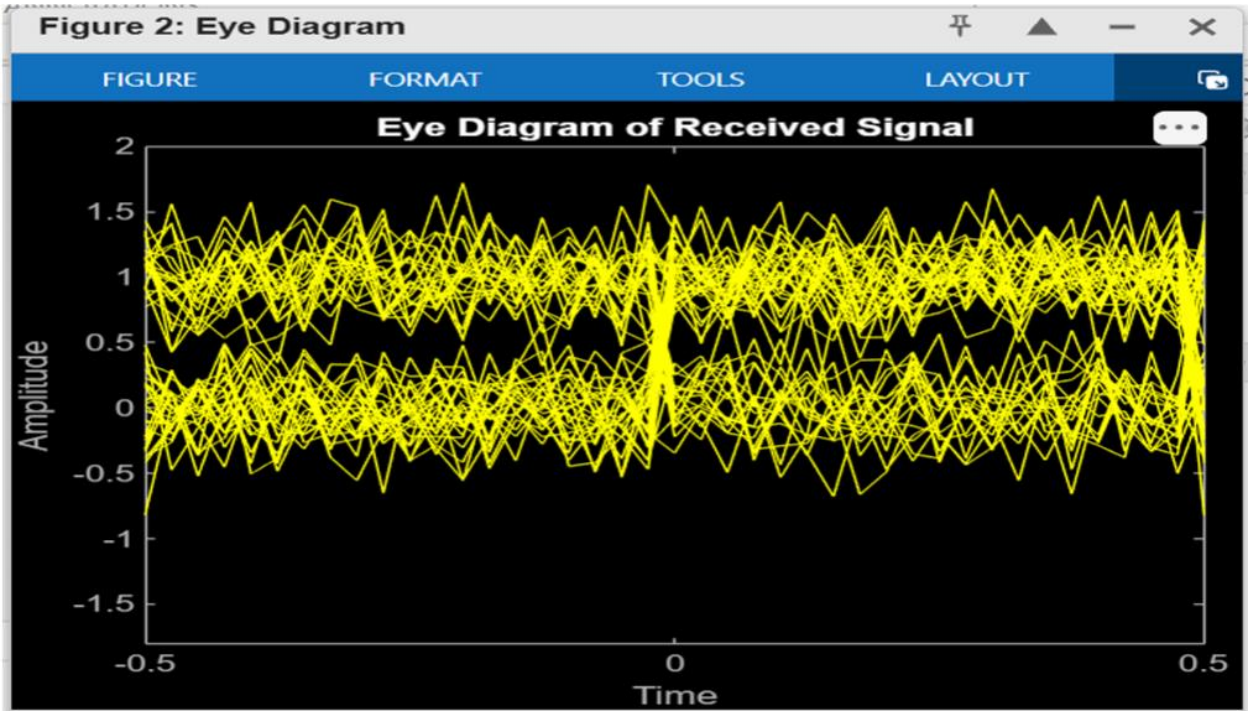
ADC EXP 17


Figure 1: NRZ Encoding and Eye Diagram
FIGURE
FORMAT
TOOLS
LAYOUT
Original Binary Data
Bit
Bit Number
Transmitted NRZ Signal
Amplitude
Time (Bit Periods)
Received Signal with AWGN (SNR = 10 dB)
Amplitude
Time (Bit Periods)
Eye Diagram of Received NRZ Signal
Amplitude
Time (Bit Periods)


Figure 2: Eye Diagram
FIGURE
FORMAT
TOOLS
LAYOUT
Eye Diagram of Received Signal
Amplitude
Time

/MATLAB Drive/ADCexp17.m

```
clc;
clear;
close all;

%% 1. PARAMETERS
N = 100;                  % Number of bits
sps = 20;                 % Samples per bit
SNR = 10;                 % Signal-to-noise ratio (dB)

%% 2. GENERATE RANDOM BINARY DATA
data = randi([0 1], 1, N);

%% 3. NRZ ENCODING (0 -> 0, 1 -> 1)
tx_signal = repelem(data, sps);

%% 4. ADD AWGN NOISE (WITHOUT awgn())
signalPower = mean(tx_signal.^2);
noisePower = signalPower / (10^(SNR/10));
noise = sqrt(noisePower) * randn(size(tx_signal));
rx_signal = tx_signal + noise;

%% 5. TIME AXIS
t = (0:length(tx_signal)-1)/sps;

%% 6. PLOT RESULTS
figure('Name','NRZ Encoding and Eye Diagram');

% Original Binary Data
subplot(4,1,1);
stairs(0:N-1, data, 'LineWidth', 1.5);
grid on;
title('Original Binary Data');
xlabel('Bit Number');
ylabel('Bit');
ylim([-0.2 1.2]);

% Transmitted NRZ Signal
subplot(4,1,2);
plot(t, tx_signal, 'LineWidth', 1.2);
grid on;
title('Transmitted NRZ Signal');
xlabel('Time (Bit Periods)');
ylabel('Amplitude');
ylim([-0.5 1.5]);

```

```
% Received Signal with AWGN
subplot(4,1,3);
plot(t, rx_signal, 'r');
grid on;
title(['Received Signal with AWGN (SNR = ', num2str(SNR), ' dB)']);
xlabel('Time (Bit Periods)');
ylabel('Amplitude');

% Eye Diagram (without eyediagram())
subplot(4,1,4);
hold on;

eyeLength = 2 * sps;
numSegments = floor(length(rx_signal)/sps) - 1;

for k = 1:numSegments
    idx = (k-1)*sps + (1:eyeLength);
    if idx(end) <= length(rx_signal)
        plot((0:eyeLength-1)/sps, rx_signal(idx), 'b');
    end
end

grid on;
title('Eye Diagram of Received NRZ Signal');
xlabel('Time (Bit Periods)');
ylabel('Amplitude');
xlim([0 2]);

%% 7. COMPLETION MESSAGE
disp('NRZ encoding, AWGN transmission, and eye diagram completed.');
```